

Work Instruction — BV-12 Assembly

Manufacturing Engineering · 2024-02 · Revision C (superseded)

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1. Scope

This instruction covers the assembly of the BV-12 valve assembly from received sub-components to the point of final inspection.

Field	Entry
Instruction	WI-BV12
Revision	Revision C
Issued	2024-02
Status	Superseded
Applies to	BV-12 valve assembly

This revision is superseded. It is retained for record purposes only and shall not be worked to. Operators finding this revision at a workstation shall return it to Manufacturing Engineering.

2. Tooling

Item	Specification	Calibration
Torque wrench	Range covering the specified value	6 months
Assembly fixture	Fixture FX-BV12-02	Annual verification
Seal insertion tool	Tool ST-14	Visual check each shift
Depth gauge	Per drawing tolerance	6 months

Tooling shall be within calibration and shall carry a current calibration label. Calibration status shall be confirmed at the start of each shift and recorded on the shift log.

3. Sub-assembly

Components shall be drawn from released stock only. Clean all mating faces before assembly; faces shall be free of machining residue, handling marks and protective compound.

Step	Operation	Check
3.1	Clean mating faces	Visual
3.2	Inspect seal seating groove	Visual and gauge
3.3	Fit seal using insertion tool ST-14	Visual
3.4	Verify seal seating depth	Depth gauge
3.5	Fit internal components per drawing	Visual

The seal shall be fitted using the insertion tool. Verify seating depth after fitting. A seal seated outside the drawing tolerance shall be removed and discarded.

4. Housing assembly

The housing is secured to the valve body with the M8 housing bolts supplied in the assembly kit. Confirm that the kit is complete and that the bolts carry the correct finish.

Present the housing squarely and seat it by hand until the flange faces meet. Engage all fasteners by hand before applying torque. Fasteners shall be fitted dry.

Tighten the fasteners to 40 Nm.

Step	Operation	Record
4.1	Confirm kit complete	Build record
4.2	Seat housing, confirm flange contact	Visual
4.3	Engage all fasteners by hand	Visual
4.4	Apply specified torque	Build record

Where a fastener is found below the specified value at inspection, the assembly shall be quarantined and reported.

5. Inspection

Check	Method	Acceptance
Flange contact	Visual and feeler	No measurable gap
Fastener torque record	Build record	All entries present
Seal seating	Build record	Within drawing tolerance
External condition	Visual	Free of damage and residue

An assembly failing any check shall be quarantined and reported. Conforming assemblies shall be labelled and passed to packing.